

Notes

- This is assumed to be testing frontend components locally as the Vercel deployment and the local version may behave differently from component to component

User Story 2.4 — Sign In with UCSC Google Account

Test Scenario 1: Successful login with valid UCSC account

Preconditions: User has a valid @ucsc.edu Google account. Firebase API key is correctly configured in .env.

Steps:

- Navigate to the Login page
- Click "Sign in with Google"
- Select or enter a @ucsc.edu Google account in the OAuth popup
- Complete authentication
- Expected Result: User is redirected to the Home Dashboard. Dashboard loads with no errors or blank panels.

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Test Scenario 2: Login blocked for non-UCSC account

Preconditions: User has a personal Gmail account (e.g. user@gmail.com)

Steps:

- Navigate to the Login page
- Click "Sign in with Google"
- Select a non-@ucsc.edu account in the OAuth popup
- Expected Result: User is not redirected to the dashboard. An error message is displayed indicating that only UCSC accounts are permitted. User remains on the Login page.

Test Scenario 3: User closes OAuth popup without completing sign-in

Preconditions: None

Steps:

- Navigate to the Login page
- Click "Sign in with Google"
- Close the OAuth popup window without selecting an account
- Expected Result: User remains on the Login page. No error is thrown. The page does not crash or freeze.

Test Scenario 4: Invalid or missing Firebase API key

Preconditions: VITE_FIREBASE_API_KEY is set to an invalid value in .env

Steps:

- Navigate to the Login page
- Click "Sign in with Google"
- Expected Result: A user-facing error message is displayed (e.g. "Authentication is currently unavailable. Please contact your administrator."). The app does not show a raw Firebase error code in the browser.

Test Scenario 5: Unauthenticated user tries to access dashboard directly

Preconditions: User is not logged in

Steps:

- Paste the dashboard URL directly into the browser address bar
- Expected Result: User is immediately redirected to the Login page. The dashboard content is not shown at all.

Test Scenario 6: Previously authenticated user revisits the app

Preconditions: User has previously signed in and the session is still valid

Steps:

- Close and reopen the browser tab with the app URL
- Expected Result: User is brought directly to the Home Dashboard without being prompted to sign in again.

User Story 3.3 — View Top Items and Top External Vendors on Dashboard

Test Scenario 1: Dashboard loads top items across all datasets

Preconditions: All four datasets (Amazon, CruzBuy, OneCard, Bookstore) have been uploaded and cleaned. User is logged in.

Steps:

- Navigate to the Home Dashboard
- Observe the Top Items section
- Expected Result: A list or table of top items is displayed showing item name, quantity, and total spend. Items from all datasets are represented.

Test Scenario 2: Top external vendors are listed and ranked

Preconditions: Data contains purchases from multiple external vendors (Amazon, Staples, Safeway, etc.)

Steps:

- Navigate to the Home Dashboard
- Observe the External Vendors panel
- Expected Result: Vendors are listed in descending order by total spend. Each entry shows vendor name and total spend amount.

Test Scenario 3: Switch between dataset tabs

Preconditions: All four datasets are loaded

Steps:

- On the Home Dashboard, click the Amazon tab
- Note the items displayed
- Click the CruzBuy tab
- Note the items displayed
- Repeat for OneCard and Bookstore tabs
- Expected Result: Each tab shows dataset-specific top items. Items change when switching tabs. No tab causes a crash or blank panel.

Test Scenario 4: Dashboard behavior when one dataset has no data

Preconditions: One dataset (e.g. Bookstore) has no uploaded data; other three are populated

Steps:

- Navigate to the Home Dashboard
- Click the Bookstore tab
- Expected Result: A fallback message is shown (e.g. "No data available for this dataset."). The page does not crash and other tabs still load correctly.

Test Scenario 5: Top items reflect applied time period filter

Preconditions: Data exists across multiple years

Steps:

- On the Home Dashboard, change the time period filter to a specific quarter (e.g. Q1 2024)
- Observe the Top Items section
- Expected Result: Items and spend totals update to reflect only purchases within the selected quarter. Items that had no purchases in that period are not shown.

User Story 4.2 — Filter and Search in Dataset Explorer

Test Scenario 1: Dataset Explorer loads all records by default

Preconditions: At least one dataset has been uploaded and cleaned. User is logged in.

Steps:

- Navigate to the Dataset Explorer page
- Expected Result: A paginated table is displayed showing cleaned transaction records. Columns include transaction date, item name, merchant name, quantity, and total price.

Test Scenario 2: Filter by dataset

Preconditions: All four datasets are loaded

Steps:

- On the Dataset Explorer, select "Amazon" from the dataset filter dropdown
- Observe the table
- Expected Result: Only Amazon transaction records are shown. Records from CruzBuy, OneCard, and Bookstore are not displayed.

Test Scenario 3: Keyword search returns matching results

Preconditions: Dataset contains items with known names (e.g. "pens")

Steps:

- Type "pens" into the search bar on the Dataset Explorer
- Observe the table
- Expected Result: Only rows containing "pens" in the item name or relevant column are shown. The table updates without a full page reload.

Test Scenario 4: Filter by time period

Preconditions: Data spans multiple years

Steps:

- On the Dataset Explorer, select a specific year (e.g. 2023) from the time period filter
- Observe the table
- Expected Result: Only records with a transaction date in 2023 are shown. Records from other years are hidden.

Test Scenario 5: Search term with no matching results

Preconditions: Dataset is loaded

Steps:

- Type a nonsense string (e.g. "zzzzz") into the search bar
- Expected Result: The table shows an empty state message (e.g. "No results found."). The page does not crash or show an error.